PROJECT COLLECTION

REZA BABAPIRI

Computer Vision & AI

Image processing and semantic segmentation

1 TECHNICAL CASE STUDIES

Engineering scope / Method / Results / Visual evidence / Limitations

Technical design, coding and analysis: Reza Babapiri

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COMPUTER VISION · SEMANTIC SEGMENTATION

Moss Detection | Classical Vision & DeepLabv3+

Hybrid active contour compared with DeepLabv3+ ResNet-50

ENGINEERING QUESTION

Which method segments moss on textured walls more reliably: classical image processing or a learned pixel-level segmentation network?

ROLE & CONTRIBUTION

I developed the MATLAB processing and segmentation workflows, compared predictions and documented results for a project submitted with a second named course participant.

ENGINEERING DECISIONS

- Implemented a classical pipeline involving green-channel enhancement, CLAHE, median filtering, Canny detection, morphology and Chan–Vese active contour.
- Implemented and evaluated DeepLabv3+ with a ResNet-50 backbone, data augmentation, labeled segmentation masks and class weighting.
- Reviewed best/worst image predictions and precision, recall, F-measure and area-ratio distributions.

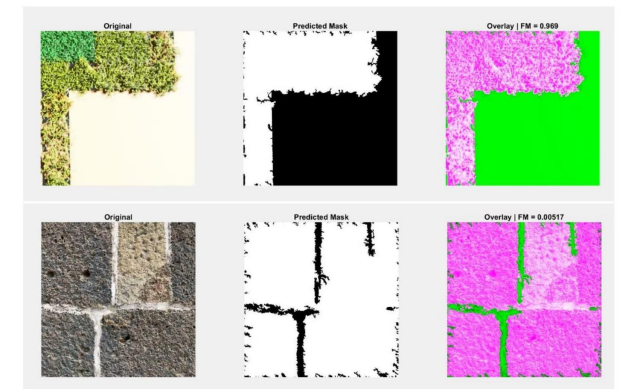
SELECTED ENGINEERING EVIDENCE

The presentation reports for the classical pipeline precision (CR) 0.833, recall (CM) 0.902 and F-measure 0.831. Deep-learning train/test distributions are plotted but no independently reproduced global benchmark is available.

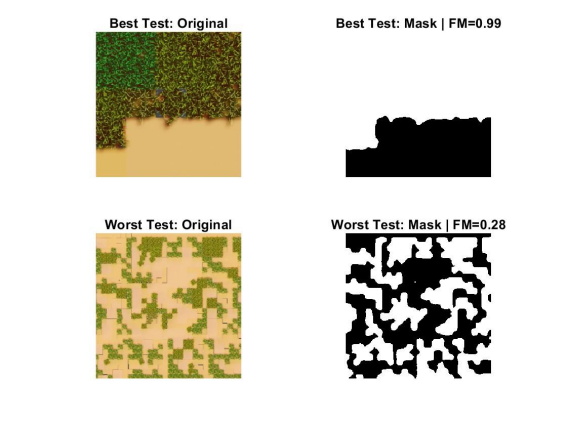
VISUAL EVIDENCE · COMPUTER VISION · SEMANTIC SEGMENTATION

Moss Detection | Classical Vision & DeepLabv3+

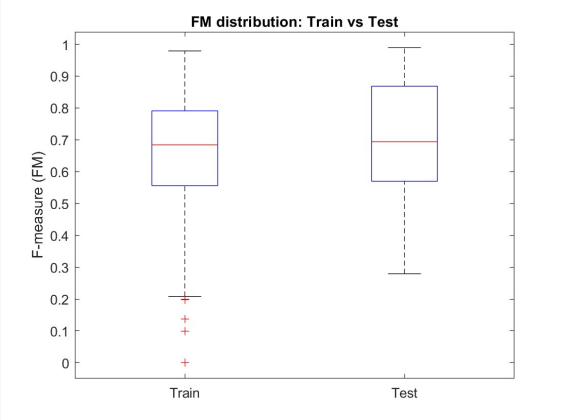
Original images, models and results from the supplied project material



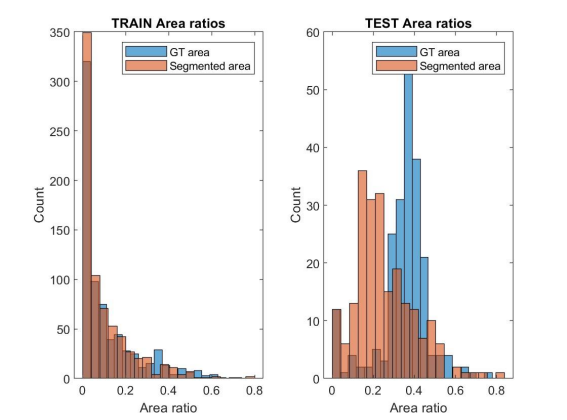
Classical segmentation: good and difficult examples for edge detection and active contours.



DeepLabv3+ qualitative best and worst test cases; predicted segmentation masks.



Training versus test F-measure distributions from the presentation.



Comparison between ground-truth and predicted vegetation areas.

RESULTS & ENGINEERING INTERPRETATION

The presentation reports for the classical pipeline precision (CR) 0.833, recall (CM) 0.902 and F-measure 0.831. Deep-learning train/test distributions are plotted but no independently reproduced global benchmark is available.